

Module-3

1. Show that $(q \rightarrow (p \wedge \sim p)) \rightarrow (r \rightarrow (p \wedge \sim p)) \equiv (r \rightarrow q)$.

$$\begin{aligned}
 (q \rightarrow (p \wedge \sim p)) \rightarrow (r \rightarrow (p \wedge \sim p)) & \equiv \sim(\sim q) \vee (\sim r) \\
 \equiv (q \rightarrow F) \rightarrow (r \rightarrow F) & \equiv q \vee (\sim r) \\
 \equiv (\sim q \vee F) \rightarrow (\sim r \vee F) & \equiv \sim r \vee q \\
 \equiv (\sim q) \rightarrow (\sim r) & \equiv (r \rightarrow q) \text{ Proved.}
 \end{aligned}$$

2. Using the truth table show that $p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$.

p	q	r	$q \vee r$	$p \rightarrow q$	$p \rightarrow r$	$p \rightarrow (q \vee r)$	$(p \rightarrow q) \vee (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	F	T	T	T	T

← Same (Proved)

3. Find the CNF of the following statement $\sim (q \vee p) \leftrightarrow (q \wedge p)$.

$$\begin{aligned}
 & \sim(q \vee p) \leftrightarrow (q \wedge p) \\
 \equiv & (\sim(q \vee p) \rightarrow (q \wedge p)) \wedge ((q \wedge p) \rightarrow \sim(q \vee p)) \\
 \equiv & (\sim\sim(q \vee p) \vee (q \wedge p)) \wedge (\sim(q \wedge p) \vee \sim(q \vee p)) \\
 \equiv & (q \vee p) \vee (q \wedge p) \wedge ((\sim q \vee \sim p) \vee (\sim q \wedge \sim p)) \\
 \equiv & (\{(q \vee p) \vee q\} \wedge \{(q \vee p) \vee p\}) \wedge (\{(\sim q \vee \sim p) \vee \sim q\} \wedge \{(\sim q \vee \sim p) \vee \sim p\}) \\
 \equiv & ((q \vee p) \wedge (q \vee p)) \wedge ((\sim p \vee \sim q) \wedge (\sim p \vee \sim q)) \\
 \equiv & (q \vee p) \wedge (\sim p \vee \sim q) \\
 \equiv & (q \vee p) \wedge \sim(p \wedge q)
 \end{aligned}$$

Ans.

4. Obtain the DNF of $\sim \{ \sim (p \leftrightarrow q) \wedge r \}$

$$\begin{aligned}
 & \sim \{ \sim (p \leftrightarrow q) \wedge r \} \\
 \equiv & \sim \sim (p \leftrightarrow q) \vee \sim r \\
 \equiv & (p \leftrightarrow q) \vee (\sim r) \\
 \equiv & \{ (p \rightarrow q) \wedge (q \rightarrow p) \} \vee (\sim r) \\
 \equiv & \{ (\sim p \vee q) \wedge (\sim q \vee p) \} \vee (\sim r) \\
 \equiv & [\{ (\sim p \vee q) \wedge \sim q \} \vee \{ (\sim p \vee q) \wedge p \}] \vee (\sim r) \\
 \equiv & [\{ (\sim p \wedge \sim q) \vee (q \wedge \sim q) \} \vee \{ (\sim p \wedge p) \vee (q \wedge p) \}] \vee (\sim r) \\
 \equiv & [\{ (\sim p \wedge \sim q) \vee F \} \vee \{ F \vee (q \wedge p) \}] \vee (\sim r) \\
 \equiv & \{ (\sim p \wedge \sim q) \vee (p \wedge q) \} \vee (\sim r) \\
 \equiv & (\sim p \wedge \sim q) \vee (p \wedge q) \vee (\sim r) \quad \underline{\text{Ans.}}
 \end{aligned}$$

5. Prove that the following equivalence $(p \wedge r) \vee (q \wedge r) \vee (\sim p \wedge (\sim q \wedge r)) \equiv r$.

$$\begin{aligned}
 & (p \wedge r) \vee (q \wedge r) \vee (\sim p \wedge (\sim q \wedge r)) \\
 \equiv & (p \wedge r) \vee (q \wedge r) \vee (\sim p \wedge \sim q \wedge r) \\
 \equiv & (p \wedge r) \vee (q \wedge r) \vee ([\sim p \wedge \sim q] \wedge r) \\
 \equiv & \{ (p \vee q) \wedge r \} \vee \{ (\sim p \wedge \sim q) \wedge r \} \\
 \equiv & [(p \vee q) \vee (\sim p \wedge \sim q)] \vee r \\
 \equiv & [(p \vee q) \vee \sim(p \vee q)] \vee r \equiv T \vee r = r \quad \underline{\text{Proved}}
 \end{aligned}$$

6. Prove that the following equivalence $(q \wedge \sim (p \wedge q)) \vee (p \wedge (\sim p \vee q)) \equiv q$.

$$\begin{aligned}
 & [q \wedge \sim(p \wedge q)] \vee [p \wedge (\sim p \vee q)] \\
 \equiv & (q \wedge (\sim p \vee \sim q)) \vee (p \wedge (\sim p \vee q)) \\
 \equiv & \{ (q \wedge \sim p) \vee (q \wedge \sim q) \} \vee \{ (p \wedge \sim p) \vee (p \wedge q) \} \\
 \equiv & (q \wedge \sim p) \vee F \vee (F \vee (p \wedge q)) \\
 \equiv & (q \wedge \sim p) \vee (p \wedge q) \\
 \equiv & (q \wedge \sim p) \vee (q \wedge p) \equiv q \wedge (\sim p \vee p) \equiv q \wedge T = q \quad \underline{\text{Proved}}
 \end{aligned}$$

7. Construct truth table and determine whether the following proposition is tautology or contradiction $[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	$(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)$	$\{[(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	F	T
T	F	T	T	F	T	F	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

(all are T)

Tautology

8. Find the truth table of $[(p \wedge q) \vee (\sim r)] \leftrightarrow p$.

p	q	r	$\sim r$	$p \wedge q$	$(p \wedge q) \vee (\sim r)$	$\{(p \wedge q) \vee (\sim r)\} \leftrightarrow p$
T	T	T	F	T	T	T
T	T	F	T	T	T	T
T	F	T	F	F	F	F
T	F	F	T	F	T	T
F	T	T	F	F	F	T
F	T	F	T	F	T	F
F	F	T	F	F	F	T
F	F	F	T	F	T	F

Contingency

9. Prove that $\sim(p \vee q) \vee (\sim p \wedge q) \vee p$ is a tautology.

p	q	$\sim p$	$(p \vee q)$	$\sim(p \vee q)$	$(\sim p \wedge q)$	$\sim(p \vee q) \vee (\sim p \wedge q) \vee p$
T	T	F	T	F	F	T
T	F	T	T	F	T	T
F	T	F	T	F	T	T
F	F	T	F	T	F	T

(all are T)

Tautology proved

10. Prove the following equivalence: $\sim(p \rightarrow q) \equiv p \wedge \sim q$

$$\begin{aligned} & \sim(p \rightarrow q) \\ \equiv & \sim(\sim p \vee q) \equiv \sim\sim p \wedge \sim q \equiv \boxed{p \wedge \sim q} \quad \text{Proved} \end{aligned}$$

11. Show that $(p \wedge q) \wedge \sim(p \vee q)$ is a contradiction.

P	q	$p \wedge q$	$p \vee q$	$\sim(p \vee q)$	$(p \wedge q) \wedge \sim(p \vee q)$
T	T	T	T	F	F
T	F	F	T	F	F
F	T	F	T	F	F
F	F	F	F	T	F

→ All F
Contradiction
Proved..

12. Define Tautology. Show that $(p \rightarrow (q \rightarrow p))$ is a tautology.

P	q	$q \rightarrow p$	$p \rightarrow (q \rightarrow p)$
T	T	T	T
T	F	T	T
F	T	F	T
F	F	T	T

→ all T
∴ Tautology
Proved.

13. Show that the two statements $(p \rightarrow q) \rightarrow q$ and $p \vee q$ are logically equivalent.

$$\begin{aligned} & (p \rightarrow q) \rightarrow q \\ \equiv & (\sim p \vee q) \rightarrow q \\ \equiv & \sim(\sim p \vee q) \vee q \\ \equiv & (\sim\sim p \wedge \sim q) \vee q \\ & \equiv (p \wedge \sim q) \vee q \\ & \equiv (p \vee q) \wedge (\sim q \vee q) \\ & \equiv (p \vee q) \wedge T \\ & \equiv \boxed{p \vee q} \quad \text{Proved.} \end{aligned}$$

14. Without truth table Prove that $p \leftrightarrow q \equiv (p \vee q) \rightarrow (p \wedge q)$

P	q	$p \vee q$	$p \wedge q$	$p \leftrightarrow q$	$(p \vee q) \rightarrow (p \wedge q)$
T	T	T	T	T	T
T	F	T	F	F	F
F	T	T	F	F	F
F	F	F	F	T	T

∴ $p \leftrightarrow q$
 $\equiv (p \vee q) \rightarrow (p \wedge q)$
(Proved)

← matched.

15. Show that the following is a tautology: $(p \wedge q) \rightarrow (p \vee q)$

p	q	$p \wedge q$	$p \vee q$	$(p \wedge q) \rightarrow (p \vee q)$
T	T	T	T	T
T	F	F	T	T
F	T	F	T	T
F	F	F	F	T

All are T

So tautology
Proved.